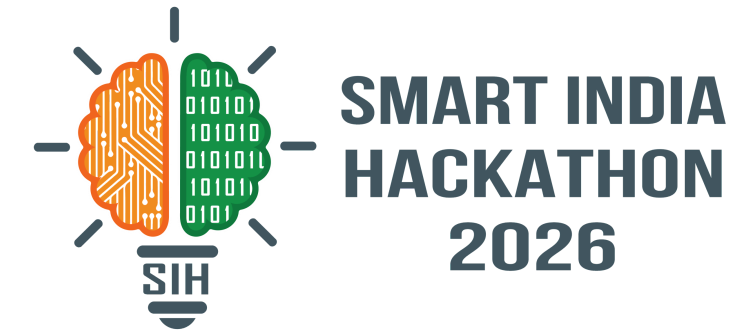
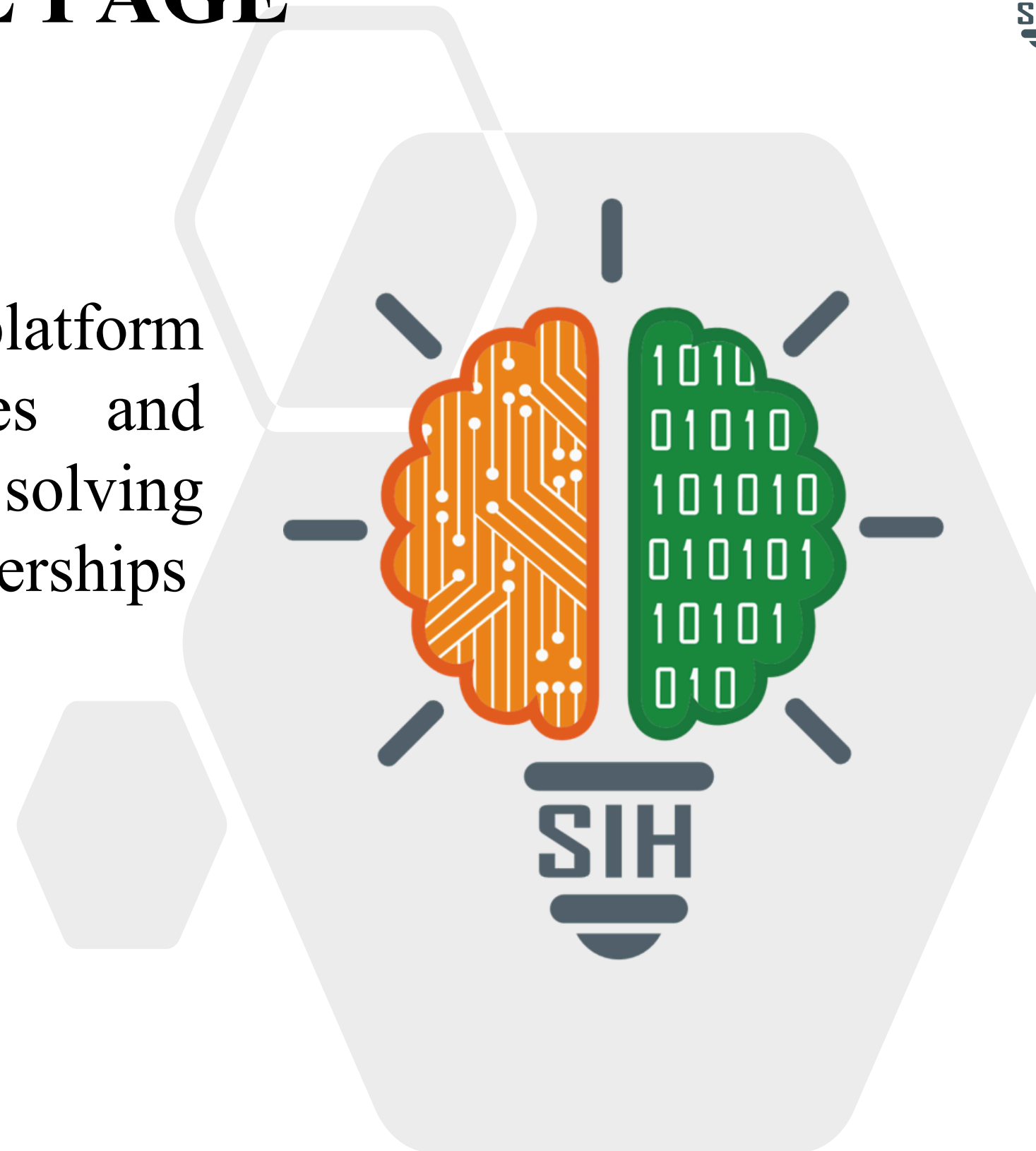


SMART INDIA HACKATHON 2026

TITLE PAGE



- **Problem Statement ID** – SIH26043
- **Problem Statement Title-** A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name :** Gauntlet



- **Proposed Solution**

- An AI-powered platform that matches citizen-reported civic/disaster problems to the university departments, industry CSR programs, or NGOs best equipped to solve them, then tracks the fix through to verified resolution.
- It closes the loop that most civic-reporting platforms leave open — connecting reported problems to real solvers via AI matching while keeping district officials in the approval loop.
- Explainable AI solver-matching combined with a built-in incentive engine (CSR compliance docs, university impact certificates, student credentials) that drives real adoption, not just reporting.

IDEA TITLE



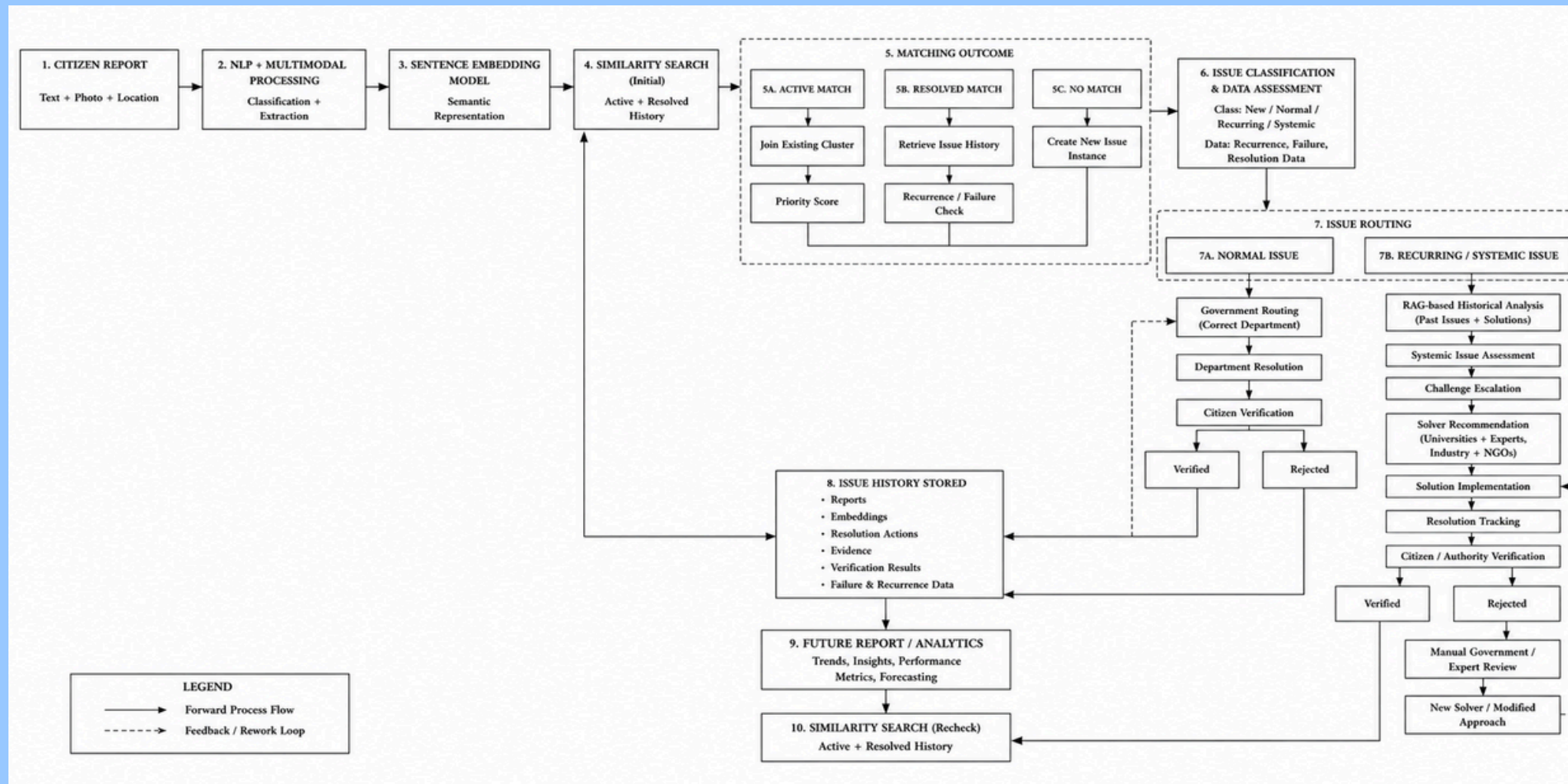
EXISTING SYSTEM

- **Individual Complaint Handling:** Each citizen complaint is treated separately.
- **Manual Categorization & Routing:** Issues are manually forwarded to departments.
- **Duplicate Complaints:** Multiple reports of the same issue create repeated tickets.
- **Limited Prioritization:** Mostly based on complaint count or basic severity.
- **No Recurrence Intelligence:** Repeated failures are not identified as systemic problems.
- **Weak Resolution Verification:** Complaint closure may not confirm actual citizen satisfaction.

PROPOSED SYSTEM

- **AI-Powered Classification:** Automatically identifies category, severity and responsible authority.
- **Smart Incident Clustering:** Combines similar nearby complaints using semantic + geo analysis.
- **Priority & Hotspot Intelligence:** Ranks incidents and visualizes critical areas on a map.
- **Cross-Issue Detection:** Connects related problems such as sewage, stagnant water and mosquitoes.
- **Recurrence Detection:** Identifies repeated failures and recommends Path 1 / Path 2 escalation.
- **Expert Collaboration:** Matches systemic challenges with suitable universities/experts.
- **Citizen Verification:** Citizens confirm whether an issue was actually resolved.

FLOW DIAGRAM



ARCHITECTURE

- **Multi-Input Reporting:** Citizens submit issues using text, photos and location.
- **AI-Based Issue Analysis:** AI understands, classifies and matches reported issues.
- **Smart Issue Management:** Duplicate reports are clustered and prioritized.
- **Government & Systemic Routing:** Issues are routed to departments or escalated based on recurrence.
- **Historical Intelligence:** RAG analyses previous issues and solutions for better decisions.
- **Solver & Resolution Management:** Connects major challenges with experts and tracks resolution.
- **Continuous Learning:** Stores issue history and improves future matching and analysis.

COMPONENTS

- **Citizen Portal:** Enables issue submission and status tracking.
- **AI Engine:** Handles classification, embeddings and similarity analysis.
- **Clustering Engine:** Detects duplicate and related issues.
- **Government Dashboard:** Supports routing, monitoring and issue management.
- **RAG & History Database:** Stores historical data and analyses previous solutions.
- **Solver Recommendation Engine:** Matches systemic challenges with suitable experts.
- **Resolution & Verification Module:** Tracks solutions and validates outcomes.

TECHNOLOGY STACK



FEASIBILITY

- Existing AI and NLP models enable automated issue understanding.
- Multimodal AI supports combined text, image and location analysis.
- Vector databases enable similarity search and duplicate detection.
- RAG enables analysis of historical issues and solutions.
- Modern web technologies support citizen and government platforms.
- Modular architecture enables gradual development and scaling.
- Human oversight improves reliability for critical decisions.

CHALLENGES & RISKS

- Incorrect or incomplete reports may affect AI accuracy.
- Limited historical data may reduce recurrence detection accuracy.
- RAG outputs may be inaccurate when retrieved data is limited or ambiguous.
- Large-scale usage may create scalability and performance challenges.
- Government integration may be complex across departments.
- Privacy and security risks exist when handling citizen data.

MITIGATION

- Input validation and human review improve the reliability of submitted reports.
- Multimodal similarity checks reduce incorrect issue matching and clustering.
- Continuous updates to verified data improve historical analysis accuracy.
- Evidence-grounded validation improves the reliability of RAG outputs.
- Structured verification and audit trails ensure accurate resolution validation.

IMPACT

- Faster Civic Issue Resolution
- Reduced Duplicate Reporting
- Improved Transparency & Accountability
- Early Detection of Systemic Problems

BENEFITS

- Streamlined Issue Management
- Data-Driven Decision Making
- Better Government–Expert Collaboration
- Continuous Learning from Past Issues

FOR INSTANCE: USER STORY

Scenario

A citizen reports a damaged road using text, a photo and location. The AI analyses the report and checks for similar active or resolved issues. A normal issue is routed to the responsible department, while a recurring problem is analysed using historical data and can be escalated to suitable experts or institutions. After resolution, the citizen verifies the outcome; rejected solutions return for rework. Verified issues are stored for future analysis and matching.

Impact

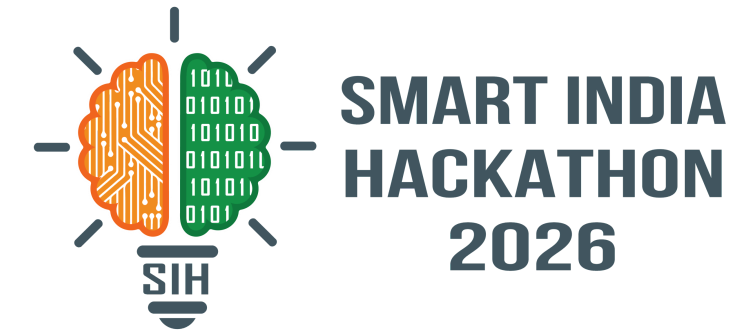
- **Faster Civic Issue Resolution:** AI-based analysis and routing reduce the time required to process and assign issues.
- **Reduced Duplicate Reporting:** Similar citizen reports are grouped into a single issue cluster.
- **Improved Transparency & Accountability:** Citizen verification confirms whether reported resolutions are successful.
- **Early Detection of Systemic Problems:** Repeated failures are identified before they become larger civic challenges.

Benefits

- **Streamlined Issue Management:** Authorities receive organised, prioritised and relevant issue information.
- **Data-Driven Decision Making:** Historical issue data supports better planning and future decisions.
- **Better Government–Expert Collaboration:** Complex challenges can be connected with universities, experts and other suitable solvers.
- **Continuous Learning from Past Issues:** Verified outcomes improve future similarity matching and historical analysis.

- Reimers & Gurevych (2019), *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks* — <https://arxiv.org/abs/1908.10084> (basis for problem/solver text-embedding matching).
- Ester, Kriegel, Sander & Xu (1996), *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise (DBSCAN)* — <https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf> (basis for duplicate/hotspot report clustering).
- CPGRAMS — Centralised Public Grievance Redress and Monitoring System, Govt. of India — <https://pgportal.gov.in/> (bootstrap grievance-data reference).

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.